The National Higher School of Artificial Intelligence
Natural Language Processing
Final Exam 22/1/2025
Duration: 2 hours

Exercise: **?? points**

Consider the following Context-Free Grammar of a minute subset of French:

S -> NP VP | S PP
NP -> Det N | Det N PP | N
VP -> V NP | V NP PP | V PP
PP -> P NP
Det -> "le" | "la" | "un" | "une"
N -> "chat" | "chien" | "fille" | "garçon" | "maison" | "parc"
V -> "voit" | "aime" | "regarde"
P -> "dans" | "avec"

1. Convert the Grammar into Chomsky Normal Form (CNF) explaining the different steps and what they produce. Respect all the conventions.

Eliminate Null Productions:
There are no null productions in this grammar.

Convert Unit Productions:
There is only one unit production; it is NP -> N and it will be replaced as follows:

NP -> "chat",
NP -> "chien", etc.

Convert rules whose right-hand side consists of more than 2 Symbols:
We need to introduce new non-terminal symbols for each of the following rules:

NP -> Det N PP
VP -> V NP PP

The conversion of these rules produces:
NP -> X_1 PP
X_1 -> Det N
VP -> X_2 PP
X_2 -> V NP

The CNF version of the grammar is now:

S -> NP VP
S -> S PP
NP -> Det N
NP -> X1 PP
X1 -> Det N
VP -> V NP
VP -> V PP
VP -> X2 PP
X2 -> V NP

PP -> P NP

NP -> "chat" | "chien" | "fille" | "garçon" | "maison" | "parc"
N -> "chat" | "chien" | "fille" | "garçon" | "maison" | "parc"
V -> "voit" | "aime" | "regarde"
P -> "dans" | "avec"
Det -> "le" | "la" | "un" | "une"

2. In the below table:
   a. Give the complete result of the CKY analysis of the sentence `"le chat voit un chien dans le parc"`. Use the same conventions that we have seen in the lectures and, in case there are different parse trees for the whole sentence, use as many versions of S at cell [0,n] as needed (i.e. S_1, S_2, S_3, etc.).
   b. Using different colours for each parse tree (NOT using any colour red, pink, or close to them), indicate the parse tree starting from the S symbol at position [0,8]. Do not colour the same arc with different colours.

Le	chat	voit	un	chien	dans	le	park
Det	NP, X_1	-	-	S	-	-	S_1, S_2
	N, NP	-	-	-	-	-	S
		V	-	X_2, VP	-	-	VP, X_2
			Det	NP, X_1	-	-	NP
				N, NP	-	-	-
					P	-	PP
						Det	NP, X_1
							N, NP

3. Let *n* be the number of words in a sentence. What are the following complexities of the CKY algorithm, giving a brief explanation for each:
   a. Worst-case time complexity.
   The time complexity of the CKY algorithm is **$O(n^3)$**, where `n` is the number of words in the sentence.
   To fill in the table any cell for a span of length *k*, we look at all possible ways of combining pairs of smaller spans (sub-spans) and there are $n - k + 1$ ways for this. Since there are n^2 cells, the worst-case time complexity of the CKY algorithm is **$O(n^3)$**.

   b. Space complexity.
   Since there are **n^2** cells in the table for a sentence of n words, even with a maximum of 2 pointers that get added to any produced non-terminal in the table cells, the space complexity of the CKY algorithm is **$O(n^2)$**.
   However, for an ambiguous grammar, the space complexity can grow significantly. Each span in the CKY table can contain multiple non-terminal symbols, and the number of entries can grow exponentially with the number of derivations.